

From per-slice lengths to a vector step

M^{-1} undoes the directional averaging

Back-projection

$$\tilde{v}_j = (1/L) \sum_{\ell} \delta_{\ell j} u_{\ell}$$

$$M = (1/L) \sum_{\ell} u_{\ell} u_{\ell}^T, \quad v_j = M^{-1} \tilde{v}_j$$

Rigid translation t : each slice sees $\delta_{\ell} = u_{\ell} \cdot t$

$$\Rightarrow \tilde{v} = M t \Rightarrow v = t$$

demo: true $t = (+2.4, -1.1)$, max atom error 0.000

Slice displacements $\delta_{\ell} u_{\ell}$ and the assembled v_j

